

1.4 Pre-Quiz: Arithmetic Sequences — Markscheme

Calculator allowed. Marks are not printed on the student sheet; they are recorded here so the set can be scored consistently. Total: 54 marks.

1. [Maximum mark: 9]

(a) The three differences $u_2 - u_1$, $u_3 - u_2$, $u_4 - u_3$:

- | | | |
|-------|-------------------|------------|
| (i) | 6, 10, 14, 18, 22 | 4, 4, 4 |
| (ii) | 3, 6, 12, 24, 48 | 3, 6, 12 |
| (iii) | 1, 4, 9, 16, 25 | 3, 5, 7 |
| (iv) | 20, 14, 8, 2, -4 | -6, -6, -6 |
| (v) | 5, 5, 5, 5, 5 | 0, 0, 0 |

A1A1

A1 for at least three of the five rows correct; A1 for all five. The negative signs in (iv) and the zeros in (v) must be present.

(b) Arithmetic: (i), (iv) and (v).

A1

$d = 4$ in (i), $d = -6$ in (iv)

A1

$d = 0$ in (v)

A1

The third A1 is for treating $d = 0$ as a legitimate common difference. A constant sequence is arithmetic; a candidate who excludes (v) earns the first two marks only.

FT: award the d marks for values that follow correctly from the candidate's own differences in (a).

(c) (ii) is not arithmetic: each term is the previous term *multiplied* by 2.

A1

(iii) is not arithmetic: these are the square numbers, $u_n = n^2$; the differences grow by 2 each time.

A1

Accept "it doubles" or "geometric with ratio 2" for (ii). Accept "squares", " $1^2, 2^2, 3^2, \dots$ ", or "the differences are the odd numbers" for (iii).

(d) "A sequence is arithmetic when the difference between consecutive terms ... **A1**

... is the same (constant) for every pair of consecutive terms."

A1

A1 for referring to the difference between consecutive (successive, neighbouring) terms, A1 for saying that difference is constant. "You add the same number each time" earns both. "It goes up by the same amount" earns both. "There is a common difference" alone earns the second mark only — it names the term without saying what it means.

2. [Maximum mark: 9]

(a) A *sequence* is an ordered list of terms. **A1**

A *series* is what you get when the terms of a sequence are added together. **A1**

Accept any wording that carries the two ideas: a sequence lists, a series sums. "A series has plus signs" earns the second mark.

(b) $4 + 9 + 14 + 19 + 24$ **A1**

$= 70$ **A1**

(c) u_5 is the fifth term of the sequence; $u_5 = 24$. **A1**

S_5 is the sum of the first five terms; $S_5 = 70$. **A1**

Each A1 needs both the meaning and the value. A value with no description, or a description with no value, earns nothing for that part.

FT: award the second A1 for the candidate's own sum from (b).

(d) 3, 7, 11, 15 — sequence $3 + 7 + 11 + 15$ — series

$u_n = 2n + 1$ — sequence $S_{20} = 430$ — series **A1A1**

A1 for any two correct, A1 for all four. $u_n = 2n + 1$ is a rule for the terms of a sequence; S_{20} is the value of a series.

(e) $S_5 = u_1 + u_2 + u_3 + u_4 + u_5$ and $S_4 = u_1 + u_2 + u_3 + u_4$, so subtracting removes the first four terms and leaves u_5 . **R1**

Accept the general statement $S_n - S_{n-1} = u_n$ explained in the same way, or a numerical demonstration on the sequence in (b): $70 - 46 = 24 = u_5$.

3. [Maximum mark: 9]

(a) $1 + 100 = 101$, $2 + 99 = 101$, $3 + 98 = 101$ — every pair has the same total, 101. **A1**

The A1 is for the observation, not for the three additions. As one number goes up by 1 the other goes down by 1, so the total does not change.

(b) The 100 numbers make $\frac{100}{2} = 50$ pairs. **M1**

$50 \times 101 = 5050$ **A1**

M1 for $100 \div 2$, or for a clear statement that the pairs run $1 + 100$ down to $50 + 51$.

(c) $u_1 = 1$, $d = 1$, $u_{100} = 100$

$S_{100} = \frac{100}{2}(1 + 100)$ **M1**

$= 50 \times 101 = 5050$ **A1**

M1 for substituting all three values into the formula; A1 for 5050 together with a statement that it agrees with (b).

FT: award the A1 for a value correctly computed from the candidate's own u_1 , d and u_{100} .

(d) Pairing the first term with the last, the second with the second-to-last, and so on gives a total of $u_1 + u_n$ every time. **R1**

There are $\frac{n}{2}$ such pairs, because n terms make n divided by 2 pairs — so $S_n = \frac{n}{2}(u_1 + u_n)$ is the number of pairs times the pair total. **R1**

The second R1 is specifically for identifying $\frac{n}{2}$ as the number of pairs. Accept the alternative explanation: write the series forwards and backwards, add the two copies to get n lots of $(u_1 + u_n)$, then halve — R1 for the doubling set-up, R1 for the halving.

(e) $\frac{500}{2} = 250$ pairs, each totalling $1 + 500 = 501$ **M1**

$250 \times 501 = 125\,250$ **A1**

Accept the formula route $S_{500} = \frac{500}{2}(1 + 500)$ for the M1.

4. [Maximum mark: 9]

(a) 21, 34, 55

A1

All three required for the mark.

(b) Differences: $1 - 1 = 0$, $2 - 1 = 1$, $3 - 2 = 1$, $5 - 3 = 2$, $8 - 5 = 3$ **M1**

The differences are not all the same, so the sequence is not arithmetic. **A1**

M1 for at least three differences computed; A1 for the conclusion linked to the differences not being constant. A bare “it is not arithmetic” earns nothing.

(c) A rule is *recursive* when each new term is built from the term or terms that came before it. **R1**

A recursive rule also needs starting value(s) — here $u_1 = u_2 = 1$ — and to reach a term you must work through all the terms before it. **R1**

Accept “the rule refers back to itself”, “the answer to one step is the input to the next step”. The second R1 is for either of the two ideas: that starting values are required, or that you cannot jump straight to u_n .

(d) Recursive: $7 \rightarrow 11 \rightarrow 15 \rightarrow 19 \rightarrow 23$, so $u_5 = 23$ **A1**

Explicit: $u_5 = 4(5) + 3 = 23$ **A1**

For u_{100} , use the explicit rule: it gives the answer in one step, $u_{100} = 4(100) + 3 = 403$, whereas the recursive rule needs 99 additions first. **R1**

The first A1 requires the intermediate terms to be shown — that is what makes it the recursive route. R1 for choosing the explicit rule with a reason about the number of steps; the value 403 is not required.

(e) $\frac{u_{10}}{u_9} = \frac{55}{34} = 1.6176\dots = 1.62$ (3 s.f.) and $\frac{u_{12}}{u_{11}} = \frac{144}{89} = 1.6180\dots = 1.62$ (3 s.f.); the ratios are settling down to about 1.618. **A1**

The mark needs both values and a comment that they are close to each other or approaching a fixed number. Naming the golden ratio is not required, but mention it when returning the work.

5. [Maximum mark: 9]

(a) The improvement is applied to its own output: the system that round n produces is the system that does the improving in round $n + 1$, exactly as u_{n+1} is defined from u_n . **R1**

Accept any answer that carries “the result of one round becomes the input to the next”. “It repeats” alone is not enough — repetition is not the same as feeding the output back in.

(b) Model A: 100, 120, 140, 160 Model B: 100, 120, 144, 172.8 **A1**

Model A is arithmetic: its differences are 20, 20, 20 — constant. Model B’s differences are 20, 24, 28.8, which are not constant. **R1**

A1 for both lists. R1 for naming Model A and giving a reason based on the differences.

(c) Model A: $c_1 = 100$, $c_{n+1} = c_n + 20$ **A1**

Model B: $c_1 = 100$, $c_{n+1} = 1.2c_n$ **A1**

Accept $c_{n+1} = c_n + 0.2c_n$ for Model B. The starting value must appear in at least one of the two rules; do not withhold both marks if it is stated once for both models.

(d) Model A: $c_{10} = 100 + 9(20) = 280$ **A1**

Model B: $c_{10} = 100(1.2)^9 = 515.97 \dots = 516$ (3 s.f.) **A1**

Accept 516, 515.98, or the full 515.978... Accept a correct value reached by stepping through all ten rounds on the GDC.

FT: award each A1 for a value that follows correctly from the candidate’s own rule in (c).

(e) Model B. **A1**

Each round adds 20% of a score that is already bigger than last round’s, so the increase itself grows — 20, then 24, then 28.8 — and the model speeds up. Model A adds the same 20 points for ever. **R1**

R1 requires the reason to be about the increase growing, not just about Model B being larger. FT: award both marks for a choice argued correctly from the candidate’s own terms in (b).

6. [Maximum mark: 9]

$$4 + 11 + 18 + \cdots + 200$$

(a) $d = 7$ A1

(b) $u_n = 4 + 7(n - 1) = 200$ M1

$7n - 3 = 200$, so $7n = 203$ and $n = 29$ A1

M1 for setting the n^{th} -term expression equal to 200, however it is written. Accept a GDC table or list used to locate 200 as the 29^{th} term.

(c) $S_{29} = \frac{29}{2}(4 + 200)$ M1

$= 29 \times 102 = 2958$ A1

Accept the other form, $S_{29} = \frac{29}{2}(2(4) + 28(7))$.

FT: award the A1 for a sum computed correctly from the candidate's own n in (b).

(d) $S_{30} = \frac{30}{2}(2(6) + 29(-2))$ M1

$= 15(12 - 58) = 15(-46) = -690$ A1

M1 for substituting $u_1 = 6$, $d = -2$, $n = 30$ into either sum formula. The negative answer is correct — the terms pass through zero and the later terms are negative.

(e) $S_{10} = \frac{10}{2}(2(5) + 9d) = 320$ M1

$5(10 + 9d) = 320$

$10 + 9d = 64$

$9d = 54$, so $d = 6$ A1

M1 for a correct equation in d ; A1 for $d = 6$. Accept the equation solved on the GDC.